Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Fox, George E Ph.D.		
Project Title:	Microorganisms in the Spacecraft Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Immunology, Infection and Hematology Team		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77204-5001	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	6	**No. of Master' Degrees:**	
No. of Master's Candidates:	6	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Willson, Richard (University of Houston)		
Grant/Contract No.:	NCC 9-58-IIH00205		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In order to better understand the microbial ecology of the spacecraft environment we are developing technology that can be used to characterize and monitor important microbial populations in the spacecraft environment with emphasis on early detection of increased levels of undesirable organisms. It is envisioned that such organisms might include some that are not normally pathogenic or problematic but become so as a result of unique characteristics of the space environment. We previously developed the theoretical underpinnings for approaches that could characterize the phylogenetic position of any bacterium that is present in a sample regardless of prior expectations of what might be found. In particular it was found that a large number of "signature sequences" whose presence corresponds with important nodes in a phylogenetic tree exist and can be identified. We are now in the process of developing and testing prototype assay systems that rely on array hybridization, mass spectrometry, or homogenous (1-step) fluorescence to detect the presence of specific signatures of interest. So as to directly test the effects of the spacecraft environmental conditions on microbial physiology at the level of gene expression we are using array hybridization to monitor the		

Task Description:	expression of each gene product in E. coli growing both aerobically and anaerobically in a rotating wall vessel (RWV) under low shear modeled microgravity (LSMMG) conditions. The specific hypothesis that is being tested is that there are specific genes that are responsive to environmental perturbations associated with the spaceflight environment, some which may play a role in mediating the pathogenicity of microorganisms. In support of these activities, we are also seeking to develop improved spacecraft-compatible methodologies for use in microbial detection. It is anticipated that samples might be taken from surfaces, water, or air. This aspect of the project includes identifying chemistries for nucleic acid isolation and detection suitable for implementation in space-borne monitoring systems. We are in particular seeking to further characterize and exploit our discovery of immobilized-metal affinity adsorption for RNA isolation and the removal of primers and dyes from samples to facilitate further analysis. Finally, we are collaborating where appropriate with other members of the Immunology, Infection and Hematology Team
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The general aim of the work being undertaken is to understand the microbial ecological and gene expression influences of the spacecraft environment and to develop monitoring technology in support of such investigations. Many of the tools being developed and the ecological insights needed are also relevant to Earth based problems. For example, bioterrorism may involve unanticipated or previously unknown bacteria. Our efforts to develop rapid methods of characterizing such organisms could be of great value in such a case. More generally, microbial identification and monitoring technologies are of considerable practical utility in medicine and microbial ecology. The ability to rapidly and simply process nucleic acid samples which is needed in the space environment may lead to improvements in Earth based microbiological methodology as well, as evidenced by the commercial interest in these NSBRI-derived technologies.
Task Progress:	[Editor's Note 2/7/2011: information from NSBRI site: see also http://www.nsbri.org/ ; project previously approved in March 2006 with a "No progress report received" note] Our work in the past year was focused on four major project areas. We have completed development of a set of 16S rRNA targeted probes that will indicate the presence of major problem and indicator bacteria in flight samples. These probes can be used with similar efficiency under a standard set of operating conditions. The utility of these probes was demonstrated on several samples isolated from various systems including Mars and Moon soil simulants. We also completed the development of a computational algorithm that allows identification of short 16S rRNA subsequences that are highly characteristic of various phylogenetic groupings. In principle, an array of appropriately designed probes based on these signature sequences could be used to determine the genetic affinity (nearest known relatives) in the absence of any prior knowledge of what the problematic organism might be. In practice, the signature sequences themselves tend to be short (15 nucleotides or less) and hence not ideal for use in arrays. We are currently developing an extension to the original algorithm that will allow us to identify very long (30-60 nucleotides) signature fragments whose sequences, despite occasional mismatches, are highly characteristic of various phylogenetic groupings. An effective set of probes may have utility in a large variety of formats. It is likely that actual implementation in space flight will be driven by mission instrumentation capabilities. Although there is considerable interested in development of array-based instruments other approaches may be preferable, especially for International Space Station applications. We therefore have focused attention on several alternative formats as well. In this regard, several probes for organisms of primary interest have been successfully implemented in a molecular beacon format. Homogenous solution assays of this type would require minimal sample processing and could be readily conducted by astronauts in flight with results signaled by the presence or absence of color changes. It was found that "red-shifted" beacons have minimal contributions from sample autofluorescence. In addition, we demonstrated the potential utility of fluorescent nucleotides such as 2-aminopurine in molecular beacon applications. Regardless of the assay system ultimately chosen, rapid and simplified systems for sample processing in space will be required. During the past year, we provided further evidence that RNA/DNA purification using compaction agents eliminates the need for preprocessing steps and that the same agents can be used to enhance the adsorption capacity of anion exchangers. Also in the past year we developed further evidence that immobilized metal affinity chromatography (IMAC) which is widely used with proteins is also effective with nucleic acids. Most recently, we have begun to look at novel ways of obtaining rapid final purification of specific RNAs such as 16S rRNA. There is preliminary evidence that the microgravity environment seen in space affects bacteria in non-obvious ways with such possible outcomes as altered drug resistance or pathogenicity. In order to explore this possibility further, we are examining the response of E. coli cells grown in simulated microgravity. In order to do this, we are using modern proteomics technology to examine the expression levels of each and every gene in E. coli when cells are grown in a low shear modeled microgravity (LSMGG) environment. Various kinetic controls have been completed and initial hybridizations with organisms grown under simulated microgravity have been made. These initial studies point to several interesting clusters of genes that appear to be part of a specific response to the LSMMG.
Bibliography Type:	Description: (Last Updated: 10/31/2019)
Articles in Peer-reviewed Journals	Kanakaraj I, Jewell DL, Murphy JC, Fox GE, Willson RC. "Removal of PCR error products and unincorporated primers by metal-chelate affinity chromatography." PLoS One. 2011 Jan 14;6(1):e14512. PMID: 21264292 , Jan-2011